WORKING PAPER · VERSION 0.3

zOS – Zero OS

A minimal, verified, real-time foundation for AI agents

20 October 2026

Abstract. This paper proposes zOS, the zero operating system: an operating system reduced to what cannot be removed. zOS consists of a bootloader, a formally verified microkernel, and the bare minimum needed to run AI agents safely and in real time. Nothing at its base is new: the seL4 microkernel with its mixed-criticality extensions, its system framework and drivers, capability-based security, open roots of trust and open agent protocols all exist as open-source work. What has to be built is the layer between agents and the world: a broker that turns human intent into bounded authority, and a safety path that keeps every motion under a verified loop. One rule binds the system: whatever is not needed to run agents is left out, and for every part that remains we state whether a guarantee depends on it and whether it is proven, tested or assumed.

The history of 1991 frames the case; three shifts visible in 2026 carry it, together with the physical turn of agents into robots, which makes real time a first principle rather than an option. zOS therefore starts where agents act physically: one robot, one task. We also state where the design can fail. The horizon is deliberately short: five years, to 2031.

Keywords: microkernel, seL4, formal verification, real time, mixed criticality, object capabilities, AI agents, robotics, open source

Prologue: Imagine it's 1991

Imagine it's 1991. We are about to release a few lines of code into a world that will be very different from today.

On 6 August 1991, a researcher at CERN posts a note about a project called WorldWideWeb to the newsgroup alt.hypertext and invites others to join (W3C 2016). Nineteen days later, on 25 August, a student at the University of Helsinki writes to comp.os.minix: *"I'm doing a (free) operating system (just a hobby, won't be big and professional like gnu) for 386(486) AT clones"* (Torvalds 1991). The first code follows in September. Neither author knows what he has started. The web will connect the planet; the hobby will end up in servers, phones and much of what lies in between.

What made that moment fertile was not brilliance alone. It was a convergence: cheap hardware nobody owned, a network that let strangers collaborate, a licence that kept the work open, and an idea small enough to start that evening.

Now imagine it's 2026. The machines that matter are no longer only desks and servers. They are agents: programs that read, decide and act on their own, and increasingly do so with arms, wheels and rotors. They all run on operating systems built for a person at a keyboard. Finding flaws in software has become cheap, because machines now do it. Proving software correct is becoming cheaper, for the same reason. Almost everything one would need to build a new foundation already exists, scattered across open-source projects, waiting to be assembled.

We are, once again, about to release a few lines of code into a world that will be very different from today. This paper asks what those lines should be. Our answer is: as few as possible.

1 The idea: zero as a design rule

zOS stands for zero OS. The name is a design rule, not a slogan: start from nothing and add only what cannot be left out. Three things

cannot be left out.

1. **A bootloader** that brings up the machine, measures what it loads, and hands over to the kernel.
2. **A microkernel** that isolates components, moves messages between them, and schedules time. Nothing else runs at its privilege level. Firmware beneath it runs higher still and is named as trusted.
3. **The bare minimum to run agents:** a way to reach a model, a way to reach tools, a broker that hands out authority, and real-time loops for anything that touches the physical world.

Everything else, from file systems and window managers to compatibility with existing software, lives outside the core or does not exist at all. Where legacy is unavoidable, it runs as a guest in an isolated virtual machine.

One rule binds the components. For every part, zOS answers two questions. **Does a guarantee depend on it?** If so, it belongs to the trusted base, even when it runs isolated: a broker that grants the wrong authority makes the system unsafe while the kernel enforces every boundary correctly. Isolation limits what a failing part can damage; it does not make the part right. **What do we know about it?** Its behaviour is proven against a specification, tested, or assumed. The aim is a trusted base that is small and, step by step, proven. What must stay assumed, such as firmware, boot code and devices that write to memory directly, is named and listed. A design that pretends otherwise is lying (Section 6).

The second principle is time. A robot arm, a drone or a vehicle does not fail gracefully when a deadline is missed. zOS therefore treats time as a resource protected by the kernel, like memory, from the first line of code (Section 4).

The third principle is that zOS invents nothing at its base. Section 5 shows that the base exists as open-source work. What has to be built is the layer that connects agents to authority and to motion, and the evidence that the assumptions of the parts fit together.

The fourth is focus. Minimality only means something against a task: what cannot be left out is decided by what the device must do. zOS therefore starts where agents act in the physical world, with one robot, one task and defined operating conditions (Section 7). A general operating system for agents is the horizon, not the starting point.

2 Why now

2.1 What 1991 teaches

On 29 January 1992, Andrew Tanenbaum declared Linux obsolete in the same newsgroup: it was monolithic and tied to one processor family (DiBona et al. 1999). Many in operating-systems research found his design argument the stronger one. Linux won anyway: on cheap hardware, under the GPL, by accepting contributions pragmatically, and by running the software people already had. zOS gives up that compatibility in its core and buys it back through Linux guests (Section 3). That is a deliberate trade, not a free one.

The microkernel did not lose on its own, though. Mach, the flagship of the 1980s, was slow. Jochen Liedtke rebuilt the idea from scratch and made inter-process communication up to 22 times faster than Mach for short messages (Liedtke 1993); against contemporary microkernels the gap was roughly a factor of 10 to 20 (Heiser and Elphinstone 2016). From this L4 line came seL4, the first general-purpose kernel with a machine-checked proof of functional correctness (Klein et al. 2009). Mach survives mainly inside Apple's hybrid XNU kernel (Apple n.d.).

The lesson is double. A good design is necessary but not sufficient: timing, licence, drivers and distribution decide. And the better design needs a moment in which its cost falls. History cannot prove that 2026 is such a moment; the three shifts below argue that it may be.

2.2 Finding flaws has become cheap

In the final of DARPA's AI Cyber Challenge in August 2025, autonomous systems found 54 of 63 planted vulnerabilities, patched 43 and discovered 18 real ones (DARPA 2025). In

April 2026, Anthropic reported that an unreleased model had found thousands of high-severity vulnerabilities, including some in every major operating system and web browser, among them a 27-year-old flaw in OpenBSD (Anthropic 2026a). A month later the company and around 50 partners reported more than 10,000 findings rated high or critical; of the 1,752 assessed further, 62 per cent were confirmed at that severity. These are the company's own figures; the bottleneck, it reported, had moved from finding to verifying, disclosing and patching (Anthropic 2026b).

When finding flaws is cheap, every line of unproven privileged code becomes a liability. The Linux kernel source approached 40 million lines in early 2025, about 25 million of them drivers (Leemhuis 2025). An installation runs only part of that tree, but all of it with full privilege. The same tools also help harden existing systems (Section 6); for systems that must not fail, we argue for less privileged code, not only more patches.

2.3 Proving correctness is becoming cheaper

seL4's first proof covered about 8,700 lines of C and took, together with design and implementation, roughly 25 person-years (Klein et al. 2009; Trustworthy Systems n.d.-a). Two developments now push that cost down. Verification tools for Rust such as Verus check systems code 3 to 61 times faster than earlier tools (Lattuada et al. 2024). And language models, paired with a proof checker that rejects every wrong step, generate proofs: a neuro-symbolic method proves 77.6 per cent of the seL4 theorems in the FVEL benchmark, though only about 20 per cent of those whose proofs exceed ten lines (He et al. 2026). Short proofs are getting cheap; long ones, and the invariants behind them, are not yet.

2.4 Agents act, and increasingly in the physical world

Classical operating systems grant a process the full authority of its user. Norm Hardy described in 1988 why this fails: a program acting on behalf of one party misuses authority granted by another, a *confused deputy* (Hardy 1988). Indirect prompt injection is the same pattern: instructions hidden in data steer an agent that acts with its user's rights (Greshake et al. 2023). Heuristic defences do not hold: adaptive attacks broke 12 published defences, most of them with success rates above 90 per cent (Nasr et al. 2026). The structural answer is old: authority must be passed explicitly, as capabilities (Dennis and Van Horn 1966; Miller 2006).

At the same time, agents are acquiring bodies. Robot stacks now separate a reasoning model that plans and calls tools from an action model that drives the motors (Google 2025). Action models emit motor commands in chunks so that a robot keeps moving at 50 Hz while the slower model thinks (Black et al. 2025). And a project for US Army unmanned ground vehicles has built proofs of concept that run ROS 2, the common robot middleware, on seL4 (Lide 2026). For such systems a missed deadline is not a slow web page. It is a dropped load or a collision.

3 Anatomy of zOS

zOS has six layers. Read from the bottom, each one exists only because the one above cannot work without it.

Hardware and root of trust. zOS needs memory management, an IOMMU for devices that write to memory directly, and a root of trust that measures the first code to run. An open option exists: OpenTitan, an open-source silicon root of trust, reached its first production chip in 2025 (CNX Software 2025). Where available, CHERI adds unforgeable pointers in hardware; it is being standardised for RISC-V (Moore 2026). Firmware runs here too, and with more privilege than the kernel: on RISC-V, seL4 Microkit relies on SBI firmware such as OpenSBI in machine mode (seL4 Microkit n.d.), on Arm the equivalent is trusted firmware at the highest exception level. This layer belongs to the trusted base and is *assumed*. zOS states a hardware profile instead of assuming any hardware will do.

Bootloader. It measures and loads one signed, static system image and then gets out

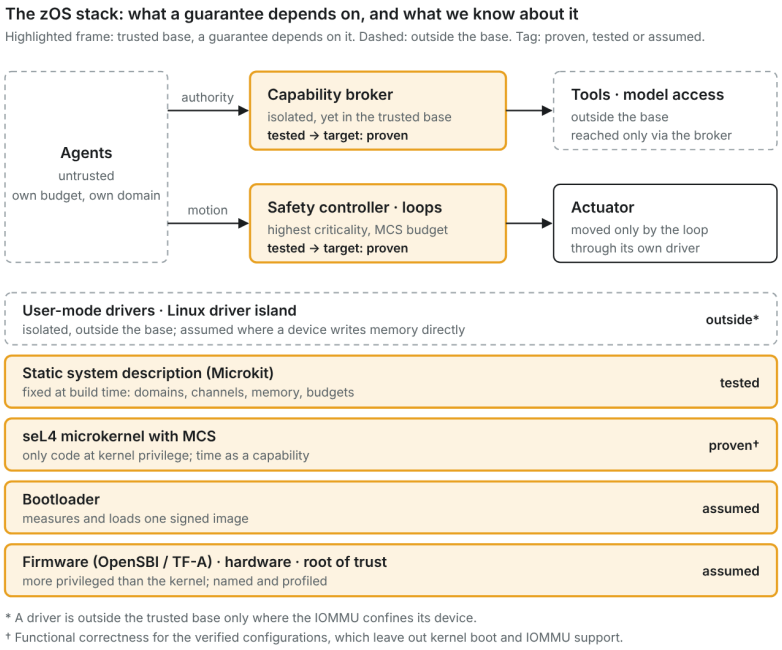


Figure 1. The zOS stack. A gold frame marks the trusted base: every part a guarantee depends on, isolated or not. The tag says what we know about it. Authority runs from the agent through the broker, motion through the safety controller; there is no third path.

of the way. It holds no policy and stays resident for nothing. Boot code lies outside seL4's verified configurations (seL4 Foundation n.d.-a), so this layer, too, is trusted base and *assumed*, and kept small for that reason.

Microkernel. seL4 with its mixed-criticality (MCS) extensions. MCS treats time, like memory, as a resource protected by capabilities (Heiser 2026). Its C-level verification is complete for RISC-V; AArch64 is targeted for the third quarter of 2027 (seL4 Foundation n.d.-b). It is the only code at the kernel's privilege level, and its functional correctness is *proven* for the verified configurations, which leave out kernel boot and IOMMU support (seL4 Foundation n.d.-a).

Static system. The whole system is declared in one description: which isolated programs exist (protection domains), which may talk (channels) and which memory they share. A tool turns that description into the bootable image (seL4 Microkit n.d.). Nothing is loaded into the core at run time. A system whose structure is fixed at build time can be checked, scheduled and audited before it ever runs.

Drivers. Drivers run in user mode, each with access only to its own device registers (Parker 2022). Legacy drivers, above all for graphics and AI accelerators, run inside a Linux virtual machine that serves as a driver island (Heiser et al. 2025). These parts are isolated as far as memory goes and stay outside the trusted base. Wherever a device can write to memory directly and the IOMMU configuration is not proven, the driver joins the trusted base as *assumed* and is listed as such. The island sits on the agent's side: neither the authority path nor the motion path runs through it. If it fails, usefulness and confidentiality suffer, not physical safety; confidentiality does, because local inference sends the model's context through it.

The agent minimum. Four components, no more:

1. **Model access:** local inference on an accelerator, reached through the driver island, or remote inference over the network.
2. **Tool access:** an open protocol for connecting agents to tools. The Model Context Protocol has been governed by the Agentic

AI Foundation under the Linux Foundation since December 2025 (Anthropic 2025).

3. **Capability broker:** the only component that hands out authority. It grants capabilities that are attenuated, time-limited and revocable. It, not the agent, words every request a human sees. It runs isolated, yet it belongs to the trusted base: its decisions are the guarantee. It is *tested* first; the target is *proven* (Section 7).
4. **Agents:** untrusted tenants, each in its own protection domain, with a CPU budget enforced by the kernel's scheduling contexts. No guarantee depends on them.

Next to the agents, and above them in criticality, sit the **real-time loops**: control, safety monitoring and actuation. Section 4 explains why they, not the agents, have the last word.

A guarantee depends on every framed part: firmware and hardware, bootloader, kernel, static system, broker and safety controller. Agents, model access and tool access stay outside that base. Two paths cross the figure, and there is no third: authority runs from the agent through the broker, motion runs through the safety controller.

4 Real time throughout

Agents are slow and fallible. Control loops are fast and must not fail. zOS builds on that asymmetry instead of hiding it.

Time is a protected resource. In seL4 MCS, CPU time is handed out through capabilities, with budgets the kernel enforces (Heiser 2026). An agent that loops, stalls or is hijacked exhausts its own budget, not the budget of the arm controller. Memory isolation keeps a compromised agent from writing into the loop; time isolation keeps it from starving it. A CPU budget is not yet an end-to-end guarantee: memory contention, interrupts, devices and communication also decide whether a loop meets its deadline, and each needs its own bound.

The agent proposes, the verified loop disposes. Robotics described the structural pattern twenty-five years ago. In the Simplex architecture, a complex high-performance controller runs the system as long as the plant stays inside a safe envelope; a simple, high-assurance controller takes over before it leaves the region from which it can still recover (Sha 2001). zOS maps this onto its layers. The agent and its action model play the complex controller: powerful, unproven, isolated, lower in criticality. The safety controller and the switching logic play the simple one: small, at the highest criticality, and candidates for proof. The agent may suggest any motion. Only the loop moves the motor.

Three goals, not one. Security: the agent cannot exceed its authority. Safety: the system violates no safety condition under stated assumptions. Usefulness: the task still gets done. None follows from the others; a controller that stops at every doubt is safe and useless. What it can reliably detect, for one robot and one task, is most of the engineering.

It has worked under attack. In DARPA's HACMS programme, the mission computer of Boeing's Unmanned Little Bird helicopter was rebuilt on seL4. In flight tests in 2017 the helicopter came under cyberattack mid-flight; compromised subsystems could not affect safe operation, and a red team given root access to the camera's virtual machine stayed confined to it (Klein et al. 2018).

Where real time is not yet provable. A real-time guarantee needs a bound on the kernel's worst-case execution time. seL4 has had a sound and complete WCET analysis, but only for the ARM11 processor. The work is archived because Arm stopped publishing worst-case instruction latencies for its out-of-order cores; open RISC-V cores such as CVA6 are named as the way forward (Trustworthy Systems n.d.-b). Protection against timing side channels points the same way: it needs a hardware instruction that so far exists only as a prototype on CVA6 (Wistoff et al. 2022). Provable schedulability for MCS remains ongoing research (Heiser 2026).

The consequence is concrete. The first zOS should run on an open RISC-V core, because only there is a proof of its timing within reach.

5 Nothing to invent at the base

The base of zOS exists today as open-source work. Table 1 lists it with its state in 2026. The list mixes running software, hardware designs, protocols, tools and patterns, and the difference matters: a proof tool is not a proven broker, and a safety pattern is not a safety controller for a given robot.

The base exists; its safe integration is a substantial engineering and research task, and it carries the contribution of zOS. The **broker** must bind each permitted action to recipients, data and parameters, enforce that at every tool call, map kernel capabilities onto external credentials, define revocation for actions under way, and word approval requests in a form the agent cannot imitate. The **tool adapters** must leave no path around it, including the context sent to remote models. The **safety path** must be specified for a concrete robot. All three need evidence that their assumptions match the parts beneath.

Others are building this. LionsOS, from the group behind seL4, already composes seL4, Microkit and user-mode drivers into a static, modular system for cyber-physical use; its driver verification is under way (Trustworthy Systems 2026). zOS should start from it, not beside it. Agent operating systems on seL4 are appearing too: agentOS gives agents capabilities for tools, models and memory and lets them replace system services at run time (Hubbard 2026); Cohesix coordinates edge AI fleets (Bower 2026); Google's KataOS paired seL4 with Rust for machine-learning devices (Google 2022). None of the three names real-time control as its goal, and services that agents replace at run time, scheduling among them, rule out what a control loop needs. zOS adds the authority layer from intent to actuator, on a static, real-time base.

zOS builds entirely on open source, and we cannot say what rules open source will follow by 2031: licences written for human authors meet machine-written code, and maintainers face a flood of machine-found flaws. The licence will matter, as it did for MINIX inside Intel's Management Engine (Tanenbaum 2017). Who maintains the integration and its proofs over years matters at least as much.

6 What zOS is not, and where it can fail

An earlier draft of this design was attacked by seven AI-assisted red-team reviews, each told to tear apart one aspect: formal methods, hardware, exploitation, agent security, supply chain, systems and facts. Their findings are published with this paper. What follows is what survived as limitation.

zOS does not solve prompt injection. Capabilities limit how far a deceived agent can reach; they do not stop the deception. An injected agent can simply ask for more authority, and humans are poor gatekeepers: in one study only about 17 per cent of Android users paid attention to permissions at install time, and 3 per cent understood them (Felt et al. 2012). Authority is also not information flow. Data can leak through channels an agent is allowed to use, which is why capability systems for agents add information-flow control (Costa et al. 2025). zOS shrinks the blast radius. It does not make agents trustworthy.

The trusted class is larger than one would like. Firmware below the kernel, from management engines to baseboard controllers, has full access to memory and cannot be confined by any kernel. Devices that write to memory directly have defeated IOMMU protection in Linux, macOS and FreeBSD (Markettos et al. 2019). And GPUs, the hardware agents depend on most, have leaked memory between processes: one flaw exposed about 181 MB per query of a running language model (Trail of Bits 2024). zOS lists these parts as trusted and demands a hardware profile. It cannot prove them.

A proof is only as good as its seams and its specification. In three formally verified distributed systems, 11 of 16 bugs found lay in the layer between verified and unverified code, two in the specifications (Fonseca et al. 2017). zOS, built from many isolated parts, consists largely of such seams. Its consequence is a rule: the two critical paths get as few seams as possible, and each seam on them

Component	Existing work	State in 2026
Root of trust	OpenTitan	First production chip announced in 2025 (CNX Software 2025)
Capability hardware	CHERI, CHERIoT	Standardisation for RISC-V under way (Moore 2026); CHERIoT pairs CHERI with a clean-slate embedded OS (Amar et al. 2023)
Microkernel	seL4 with MCS	Functional correctness proven (Klein et al. 2009); MCS verified at C level on RISC-V, AArch64 targeted for Q3 2027 (seL4 Foundation n.d.-b)
System composition	seL4 Microkit	Available for use, under active development (seL4 Microkit n.d.)
Drivers	seL4 Device Driver Framework, LionsOS	Modular OS on seL4 that outperforms Linux on system-call-intensive loads; legacy drivers in VMs (Heiser et al. 2025)
Verified components	Verus for Rust	Verifies systems code 3 to 61 times faster than prior tools (Lattuada et al. 2024)
Proof automation	AutoVerus; neuro-symbolic proof search	Over 90 % of 150 Rust proof tasks (Yang et al. 2025); 77.6 % of seL4 theorems in FVEL (He et al. 2026)
Real-time safety pattern	Simplex architecture	Established since 2001 (Sha 2001)
Robot middleware	ROS 2 on seL4	Proofs of concept for Army ground vehicles (Lide 2026)
Agent–tool protocol	Model Context Protocol	Open, under the Linux Foundation since December 2025 (Anthropic 2025)
Agent authority	CaMeL, FIDES	Capability and information-flow enforcement for agents; CaMeL solves 77 % of tasks with provable security versus 84 % undefended (Debenedetti et al. 2025; Costa et al. 2025)

Table 1. Nothing to invent: the open-source parts of zOS and their state in 2026.

is specified as an interface that is checked when the system image is built. Multicore verification of seL4 is scheduled for 2028 (seL4 Foundation n.d.-b), so a proven zOS starts on a single core per kernel instance.

Hardening may win instead. Writing new code in a memory-safe language, without rewriting the old, works: in Android, memory-safety flaws fell below 20 per cent of all vulnerabilities in 2025 for the first time (Vander Stoep 2025). That is exactly how Linux won in 1991: good enough, incremental, invisible to users. Linux restricts processes with Landlock (Linux kernel n.d.-a) and supports real time with PREEMPT_RT (Linux kernel n.d.-b). The obvious alternative to zOS is an agent on hardened Linux plus a separate safety controller for the motors. For the actuators that split is sound, and zOS adopts it. It leaves out the authority the agent holds over data, credentials and tools, and what it sends out, including the context shipped to remote models. A safety controller guards motion, not meaning. zOS only makes sense where hardening cannot reach: explicit authority for agents, provable timing for machines that move, and a small trusted base for systems that must not fail.

The Hurd trap. GNU Hurd, a multi-server system on Mach, never reached broad use. Its history has more than one cause, but one lesson holds for zOS: start smaller than the vision, with one board, one kernel, one agent, one loop.

7 Five years back from 2031

The path is read backwards from a deliberately modest goal: by 2031, forty years after Linux, one kind of robot doing one defined task runs on zOS from boot to actuator. Each station below is speculative, except the starting point and the verification dates the seL4 Foundation has announced. Each names its ev-

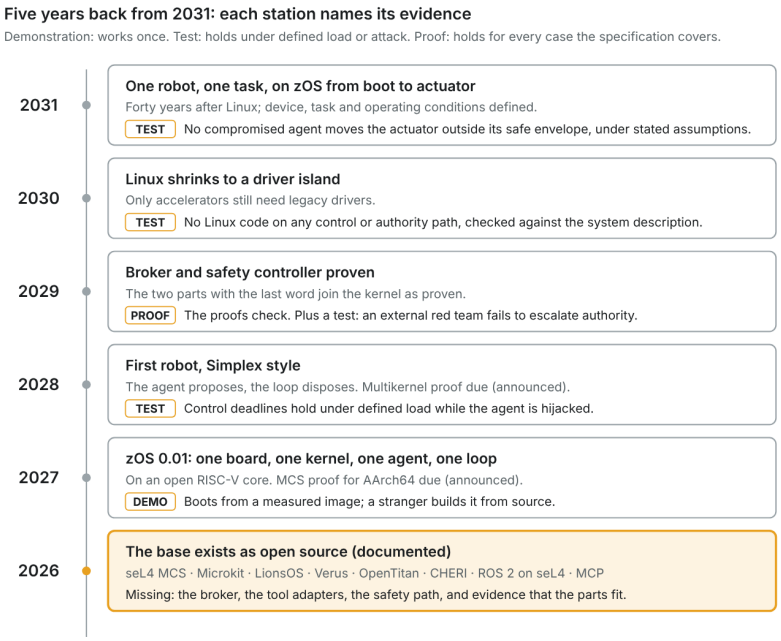


Figure 2. The path from 2026 to 2031. Each station names its evidence and whether it is a demonstration, a test or a proof; a failed check stops the path.

idence and its kind: a **demonstration** shows that something works once, a **test** that it holds under defined load or attack, a **proof** that it holds for every case its specification covers. A failed red team is evidence, not proof. A station that fails its check stops the path, which makes the plan falsifiable rather than a wish.

Read from the bottom up, each station makes the next one possible. The later stations depend on dates the seL4 Foundation has announced; if those slip, the path slips. zOS 0.01 does not wait for them: on RISC-V, with one core, the MCS proof it needs exists today. The decisive station is the first: zOS 0.01 is as small as Linux 0.01 was, and just as easy to dismiss.

A possible first robot. pib, an open-source humanoid from Nuremberg with an education kit for schools, shows what this could look like; nothing of it is agreed or built. pib runs on a Raspberry Pi. Its servos are driven by small controllers with their own microcontrollers and open firmware, and a language-model agent can move its joints through MCP tools (pib.rocks 2026; Tinkerforge n.d.). As is common in educational robots, the joint limits are checked in software on the Linux side. A zOS path would start without new hardware: limits, a speed bound and a heartbeat timeout move into the controller firmware, the loop's last word in miniature. Next comes a broker in front of the agent's tools. Then seL4, which Microkit supports on the Raspberry Pi 4 (seL4 Microkit n.d.), with pib's existing Linux as a driver island beside a native safety component.

Epilogue

In 1991, Linus Torvalds did not announce an ideal system. He announced a useful one, small enough to start, open enough to grow. zOS asks for the same modesty in a different world. Its task is not to make a free Unix for a cheap PC. Its task is to make the smallest system on which an agent can act, and move, without being trusted.

None of its base is new. Capabilities date from the 1960s, the verified kernel from the 2000s, the Simplex pattern from 2001. What is new is that the reasons they stayed in niches are falling at the same time, and that the machines which need them are about to leave the screen and enter the room.

The announcement of 1991 promised a sys-

tem that "won't be big". zOS makes the same promise. This time on purpose.

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